

Project CSE423

1. The Mechanic: Line-of-Sight Combat

- The Boss (Titan) stands in the center and rotates to face the player.
- The Boss has a "Laser Eye." It draws a thin red line to the player constantly.
- **If the line touches a Shield:** The line stops at the shield. You are safe.
- **If the line touches the Player:** The Boss creates a thick "Beam" and you lose a life.

2. The Environment: Destructible Pillars

- There are 4 Pillars in the arena.
- If a Pillar takes a Laser hit (saves you), it turns **Black**.
- If it takes a second hit, it **Vanishes**.
- *Technical Flex:* This forces the player to run from cover to cover as the arena gets destroyed.

3. The Weakness: The "Cooldown" Core

- The Boss fires for 5 seconds (Red Phase).
- Then it **Overheats** for 3 seconds (Blue Phase).
- **Visual Cue:** A Blue Sphere appears in the Boss's chest.
- **Action:** The player must press **Space** to shoot a projectile at that Blue Sphere.
- **Scoring:** 3 hits to the Core = Boss Defeated.

4. The Win Condition

- Upon defeat, the Boss falls apart (scale Y down to 0).
 - The Exit Door (a simple rectangle on the wall) rotates 90 degrees open.
 - Player walks through -> "YOU SURVIVED" text appears.
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- **Line-of-Sight Mechanic:** The Boss functions as a central turret that rotates to track the player. A ray-cast detects collisions: if the ray hits the player, they take damage; if it hits a pillar, the player is shielded.
 - **Destructible Environment:** The arena contains four pillars used for cover. These objects have health states: the first hit changes the texture color (black), and the second hit deletes the object, dynamically removing cover from the map.
 - **Boss State Machine:** The enemy relies on a timer loop. It attacks for 5 seconds (Red Phase) and then enters a vulnerable cooldown for 3 seconds (Blue Phase). The player must shoot a specific target object during the cooldown. Three hits defeat the boss.

- **Win Condition & Animation:** Upon zero health, the Boss object scales down to disappear. An exit door object rotates 90 degrees to open. Moving the player through the door triggers the "Success" UI text.